

Covid-19 Historical Time-Series US Data Analysis

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2025-04-09

Project Purpose

The purpose of this project is to analyze Covid-19 Time-Series US dataset from the John Hopkins GitHub site, a historical time-series data of covid-19 cases and deaths of all US counties, using R.

Objective

Main objective of this project is:

- Which US states are most affected by Covid-19?
- Can we predict future Covid-19 cases and deaths in US with a Linear Regression Model?

Load packages

```
library('tidyverse')
library('lubridate')
library('ggplot2')
library('dplyr')
```

Data Description

CSSE COVID19 Time Series Data

This folder contains daily time series summary tables, including confirmed, deaths and recovered. All data is read in from the daily case report. The time series tables are subject to be updated if inaccuracies are identified in our historical data.

Two time series tables are for the US confirmed cases and deaths, reported at the county level. They are named `time_series_covid19_confirmed_US.csv`, `time_series_covid19_deaths_US.csv`, respectively. These datasets have been used for this analysis.

Three time series tables are for the global confirmed cases, recovered cases and deaths. Australia, Canada and China are reported at the province/state level. Dependencies of the Netherlands, the UK, France and Denmark are listed under the province/state level. The US and other countries are at the country level. The tables are renamed `time_series_covid19_confirmed_global.csv` and `time_series_covid19_deaths_global.csv`, and `time_series_covid19_recovered_global.csv`, respectively.

Update frequency: Once a day around 23:59 (UTC).

Link: https://github.com/CSSEGISandData/COVID-19/tree/master/csse_covid_19_data/csse_covid_19_time_series

Data Load

```
# All files begin with this string.
url_in <- ("https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_co

# Vector containing four file names.
file_names <-
  c("time_series_covid19_confirmed_US.csv",
    "time_series_covid19_deaths_US.csv")

# String concatenate url_in and each of the file names.
urls <- str_c(url_in, file_names)

# Store each dataset in a variable.
US_cases <- read_csv(urls[1])
US_deaths <- read_csv(urls[2])
```

Tidy and Transform Data

- Move date to a separate row and converted into date objects.
- Remove unnecessary columns (Lat, Long_).

```
US_cases <- US_cases %>%
  pivot_longer(cols = -(UID:Combined_Key),
               names_to = "date",
               values_to = "cases") %>%
  select(Admin2:cases) %>%
  mutate(date = mdy(date)) %>%
  select (-c(Lat, Long_))
```

```
US_deaths <- US_deaths %>%
  pivot_longer(cols = -(UID:Population),
               names_to = "date",
               values_to = "deaths") %>%
  select(Admin2:deaths) %>%
  mutate(date = mdy(date)) %>%
  select (-c(Lat, Long_))
```

- Join US cases and US deaths per date.

```
US <- US_cases %>%
  full_join(US_deaths)
```

```
## Joining with 'by = join_by(Admin2, Province_State, Country_Region,
## Combined_Key, date)'
```

- glimpse of the data.

```
glimpse(US)
```

```
## Rows: 3,819,906
## Columns: 8
## $ Admin2      <chr> "Autauga", "Autauga", "Autauga", "Autauga", "Autauga", ~
## $ Province_State <chr> "Alabama", "Alabama", "Alabama", "Alabama", "Alabama", ~
## $ Country_Region <chr> "US", "US", "US", "US", "US", "US", "US", "US", "US", "~
## $ Combined_Key  <chr> "Autauga, Alabama, US", "Autauga, Alabama, US", "Autaug~
## $ date          <date> 2020-01-22, 2020-01-23, 2020-01-24, 2020-01-25, 2020-0~
## $ cases          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Population     <dbl> 55869, 55869, 55869, 55869, 55869, 55869, 55869, 55869, ~
## $ deaths         <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
```

Summary of US Data (Descriptive Statistics)

```
summary(US)
```

```
##      Admin2      Province_State      Country_Region      Combined_Key
## Length:3819906 Length:3819906 Length:3819906 Length:3819906
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##
##      date      cases      Population      deaths
## Min.   :2020-01-22 Min.   : -3073 Min.   :      0 Min.   : -82.0
## 1st Qu.:2020-11-02 1st Qu.:   330 1st Qu.:   9917 1st Qu.:   4.0
## Median :2021-08-15 Median :   2272 Median :   24892 Median :   37.0
## Mean   :2021-08-15 Mean   :  14088 Mean   :   99604 Mean   :  186.9
## 3rd Qu.:2022-05-28 3rd Qu.:   8159 3rd Qu.:   64979 3rd Qu.:  122.0
## Max.   :2023-03-09 Max.   :3710586 Max.   :10039107 Max.   :35545.0
```

Row Description (US Data)

- Each **row** in the US table is a date.

Column Description (US Data)

- **Admin2**: County name
- **Province_State**: State name
- **Country_Region**: US
- **Combined_Key**: Puts together county and state
- **date**: Date in year/month/day format
- **cases**: Total number of COVID19 cases
- **Population**: Population of the County
- **deaths**: Total number of COVID19 related deaths

Visualizations and Analysis

- For the analysis, related data frames have been created with data group by Province_State.
- New columns new_cases and new_deaths have been created to calculate daily changes.

```
# Group data by Province_State & date and add mortality rate column.
states_detail_data <- US %>%
  group_by(Province_State,date) %>%
  summarize(cases = sum(cases), deaths = sum(deaths),
            Population= sum(Population), .groups = 'drop') %>%
  mutate(new_cases = cases - lag(cases),
         new_deaths = deaths - lag(deaths)) %>%
  mutate(new_deaths = if_else(is.na(new_deaths),0,new_deaths),
         new_cases = if_else(is.na(new_cases),0,new_cases)) %>%
  select(Province_State,date, cases, deaths, Population, new_cases, new_deaths) %>%
  ungroup()
```

- US total cases and deaths data frame also have been created.
- New columns mortality_rate, new_cases and new_deaths have been created

```
# Sum all cases, deaths, and populations.
us_totals <- states_detail_data %>%
  group_by(date) %>%
  summarize(cases = sum(cases), deaths = sum(deaths)) %>%
  mutate(mortality_rate = if_else(cases==0,0,deaths/cases),
         new_cases = cases - lag(cases),
         new_deaths = deaths - lag(deaths)) %>%
  mutate(new_deaths = if_else(is.na(new_deaths),0,new_deaths),
         new_cases = if_else(is.na(new_cases),0,new_cases)) %>%
  select(date, cases, deaths, mortality_rate,new_cases,new_deaths) %>%
  ungroup()
```

- Latest data for each States extract using max date present in the data.
- New columns mortality_rate, cases_per_100 and deaths_per_100 have been created.

```
max_date <- max(US$date)

# Group data by Province_State and add mortality rate column
states_total <- states_detail_data %>%
  filter(date == max_date) %>%
  group_by(Province_State) %>%
  mutate(mortality_rate = if_else(cases==0,0,deaths/cases),
         cases_per_100 = if_else(Population==0,0,cases/Population * 100),
         deaths_per_100 = if_else(Population==0,0,deaths/Population * 100)) %>%
  select(Province_State, cases, deaths, mortality_rate, cases_per_100, deaths_per_100, Population) %>%
  ungroup()
```

- Analyze data frames to find information relevant to the question of interest.

```
# Total US cases to date and Max 1 day cases.
sum(us_totals$cases)
```

```
## [1] 53813347993
```

```
# Total US deaths to date and Max 1 day cases.  
sum(us_totals$deaths)
```

```
## [1] 713875231
```

```
# Max 1 day cases  
max(us_totals$new_cases)
```

```
## [1] 1354508
```

```
# Max 1 day deaths  
max(us_totals$new_deaths)
```

```
## [1] 4375
```

```
# Avg mortality rate:  
sum(us_totals$deaths)/sum(us_totals$cases)
```

```
## [1] 0.01326577
```

```
# Maximum new cases for 1 day  
states_detail_data %>% slice_max(new_cases, n=5)
```

```
## # A tibble: 5 x 7
```

##	Province_State	date	cases	deaths	Population	new_cases	new_deaths
##	<chr>	<date>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	California	2022-01-16	7121304	77686	39512223	207110	53
## 2	California	2022-01-09	6259048	76968	39512223	197334	14
## 3	California	2022-01-23	7863053	78538	39512223	179073	69
## 4	California	2022-01-11	6542526	77137	39512223	151840	21
## 5	Florida	2022-01-04	4360178	62541	21477737	150251	37

```
# Maximum new cases for 1 day  
states_detail_data %>% slice_max(new_deaths, n=5)
```

```
## # A tibble: 5 x 7
```

##	Province_State	date	cases	deaths	Population	new_cases	new_deaths
##	<chr>	<date>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	Missouri	2021-12-02	919712	15539	6626371	5188	2441
## 2	Tennessee	2021-12-22	1364571	20266	6829174	5873	2115
## 3	Oklahoma	2021-04-07	441906	6669	3956971	1764	1716
## 4	Florida	2021-09-16	3473873	50811	21477737	9760	1554
## 5	Florida	2021-09-02	3291225	45909	21477737	21723	1338

```
# Maximum mortality rate overall  
states_total %>% slice_max(mortality_rate,n=5)
```

```
## # A tibble: 5 x 7
##   Province_State cases deaths mortality_rate cases_per_100 deaths_per_100
##   <chr>          <dbl> <dbl>          <dbl>          <dbl>          <dbl>
## 1 Grand Princess    103     3          0.0291           0           0
## 2 Pennsylvania    3527854 50398          0.0143          27.6         0.394
## 3 Oklahoma        1290929 17972          0.0139          32.6         0.454
## 4 Georgia          3068208 42489          0.0138          28.9         0.400
## 5 Michigan         3064125 42205          0.0138          30.7         0.423
## # i 1 more variable: Population <dbl>
```

```
# Maximum cases overall
states_total %>% slice_max(cases, n=5)
```

```
## # A tibble: 5 x 7
##   Province_State cases deaths mortality_rate cases_per_100 deaths_per_100
##   <chr>          <dbl> <dbl>          <dbl>          <dbl>          <dbl>
## 1 California    12129699 101159          0.00834          30.7         0.256
## 2 Texas          8466220  93390          0.0110          29.2         0.322
## 3 Florida        7574590  86850          0.0115          35.3         0.404
## 4 New York       6794738  77157          0.0114          34.9         0.397
## 5 Illinois       4083292  41496          0.0102          32.2         0.327
## # i 1 more variable: Population <dbl>
```

```
# Maximum deaths overall
states_total %>% slice_max(deaths, n=5)
```

```
## # A tibble: 5 x 7
##   Province_State cases deaths mortality_rate cases_per_100 deaths_per_100
##   <chr>          <dbl> <dbl>          <dbl>          <dbl>          <dbl>
## 1 California    12129699 101159          0.00834          30.7         0.256
## 2 Texas          8466220  93390          0.0110          29.2         0.322
## 3 Florida        7574590  86850          0.0115          35.3         0.404
## 4 New York       6794738  77157          0.0114          34.9         0.397
## 5 Pennsylvania    3527854  50398          0.0143          27.6         0.394
## # i 1 more variable: Population <dbl>
```

```
# Maximum cases_per_100 overall
states_total %>% slice_max(cases_per_100, n=5)
```

```
## # A tibble: 5 x 7
##   Province_State cases deaths mortality_rate cases_per_100 deaths_per_100
##   <chr>          <dbl> <dbl>          <dbl>          <dbl>          <dbl>
## 1 Rhode Island    460697   3870          0.00840          43.5         0.365
## 2 Alaska          307655   1486          0.00483          41.5         0.201
## 3 Kentucky        1718471  18130          0.0106          38.5         0.406
## 4 North Dakota    286950   2470          0.00861          37.7         0.324
## 5 Guam            61027    420          0.00688          37.2         0.256
## # i 1 more variable: Population <dbl>
```

```
# Maximum deaths_per_100 overall
states_total %>% slice_max(deaths_per_100, n=5)
```

```
## # A tibble: 5 x 7
##   Province_State   cases deaths mortality_rate cases_per_100 deaths_per_100
##   <chr>          <dbl> <dbl>         <dbl>         <dbl>         <dbl>
## 1 Arizona        2443514 33102         0.0135         33.6         0.455
## 2 Oklahoma        1290929 17972         0.0139         32.6         0.454
## 3 Mississippi     990756 13370         0.0135         33.3         0.449
## 4 West Virginia   642760  7960         0.0124         35.9         0.444
## 5 New Mexico      670929  9061         0.0135         32.0         0.432
## # i 1 more variable: Population <dbl>
```

Observations:

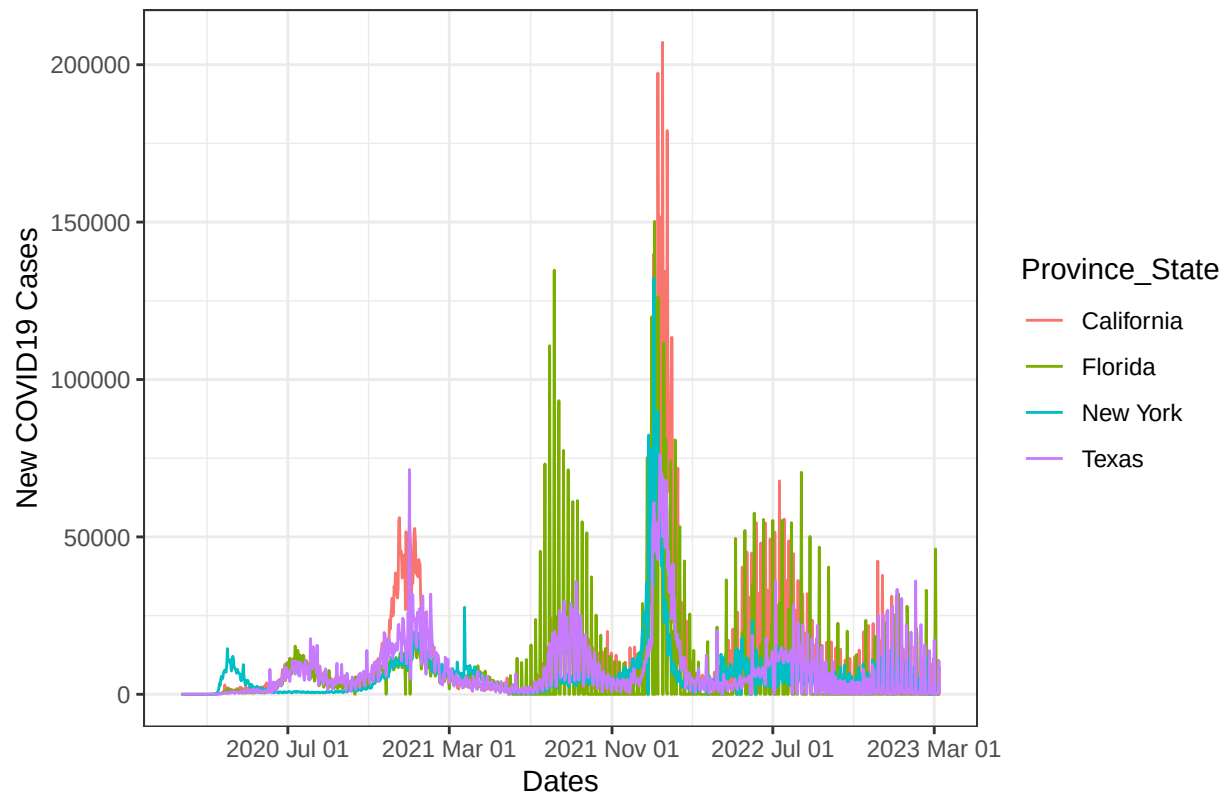
- Total COVID-19 cases for US is 53,813,347,993 whereas number of deaths are 713,875,231, showing an average mortality rate of 0.01326577.
- As per new cases for a day, maximum was reported during January, 2022 with highest as 207,110 new cases. California was the most affected state followed by Florida in this regard.
- As per new deaths for a day, maximum was reported on December, 2021. Missouri reported the highest as 2,441 deaths on 2021-12-02 followed by Tennessee with 2,115 deaths on 2021-12-22. Missouri & Tennessee was the most affected states followed by Oklahoma & Florida in this regard.
- In terms of mortality rate, the most affected states were Grand Princess, Pennsylvania, Oklahoma, Georgia & Michigan.
- In terms of total cases, the most affected states were California, Texas, Florida, New York & Illinois. However when checked by cases_per_100, it seems the most affected states were Rhode Island, Alaska, Kentucky, North Dakota & Guam.
- In terms of total deaths, the most affected states were California, Texas, Florida, New York & Pennsylvania. However when checked by deaths_per_100, it seems the most affected states were Arizona, Oklahoma, Mississippi, West Virginia & New Mexico.

By total cases and deaths, most affected states: California, Texas, Florida & New York

```
# Create a new dataframe for Iron County and add columns for daily new cases and deaths.
top_total_cases <- states_detail_data %>%
  filter(Province_State %in% c("California", "Texas", "Florida", "New York"),
         new_cases >= 0, new_deaths >= 0)

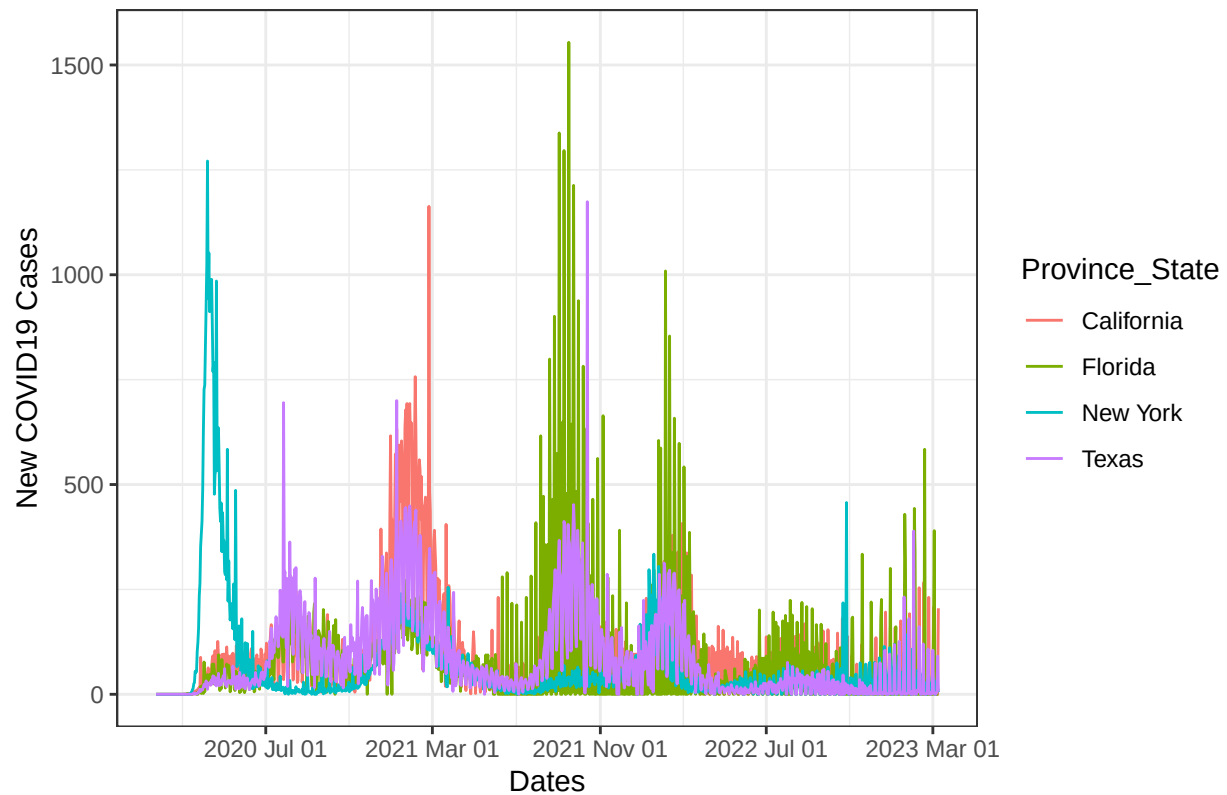
ggplot(top_total_cases, aes(x=date)) +
  geom_line(aes(y = new_cases, color=Province_State)) +
  scale_x_date(date_labels = "%Y %b %d", date_breaks = "8 month") +
  theme_bw() +
  labs(x = "Dates",
       y = "New COVID19 Cases",
       title = "Most affected states New COVID19 Cases - Time Series")
```

Most affected states New COVID19 Cases – Time Series



```
ggplot(top_total_cases, aes(x=date)) +
  geom_line(aes(y = new_deaths,color=Province_State)) +
  scale_x_date(date_labels = "%Y %b %d", date_breaks = "8 month") +
  theme_bw() +
  labs(x = "Dates",
       y = "New COVID19 Cases",
       title = "Most affected states New COVID19 Deaths - Time Series")
```


Most affected states New COVID19 Deaths – Time Series

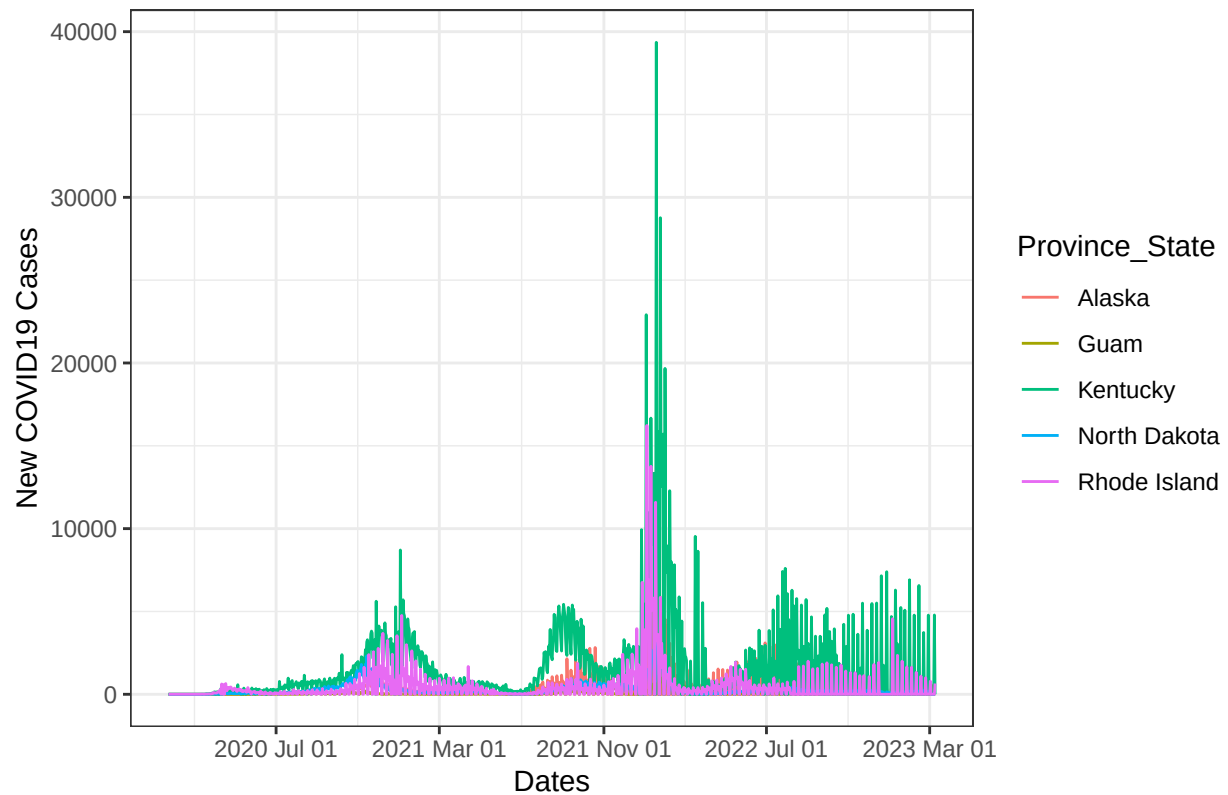


By cases_per_100, most affected states: Rhode Island, Alaska, Kentucky, North Dakota & Guam

```
# Create a new dataframe for Iron County and add columns for daily new cases and deaths.
top_states_cases <- states_detail_data %>%
  filter(Province_State %in% c("Rhode Island", "Alaska", "Kentucky", "North Dakota", "Guam"),
         new_cases >= 0, new_deaths >= 0)

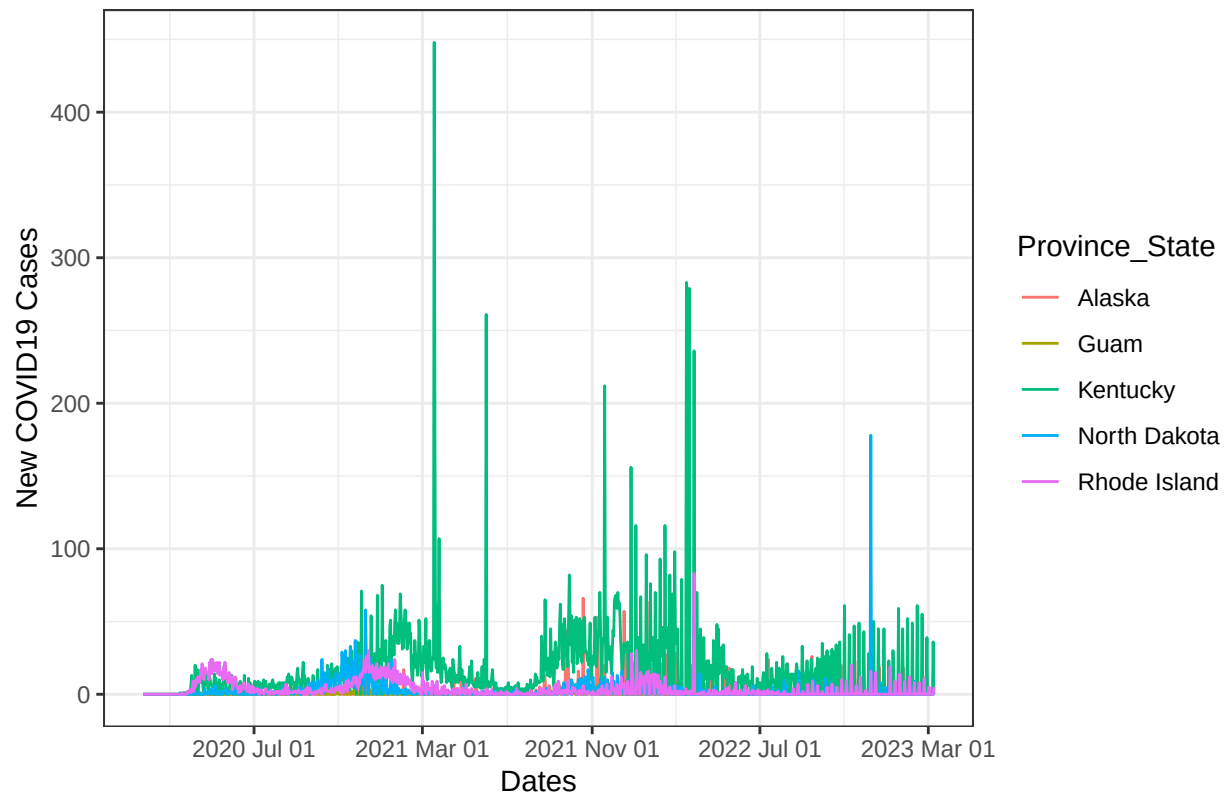
ggplot(top_states_cases, aes(x=date)) +
  geom_line(aes(y = new_cases, color=Province_State)) +
  scale_x_date(date_labels = "%Y %b %d", date_breaks = "8 month") +
  theme_bw() +
  labs(x = "Dates",
       y = "New COVID19 Cases",
       title = "Most affected states New COVID19 Cases - Time Series")
```

Most affected states New COVID19 Cases – Time Series



```
ggplot(top_states_cases, aes(x=date)) +
  geom_line(aes(y = new_deaths,color=Province_State)) +
  scale_x_date(date_labels = "%Y %b %d", date_breaks = "8 month") +
  theme_bw() +
  labs(x = "Dates",
       y = "New COVID19 Cases",
       title = "Most affected states New COVID19 Deaths - Time Series")
```

Most affected states New COVID19 Deaths – Time Series

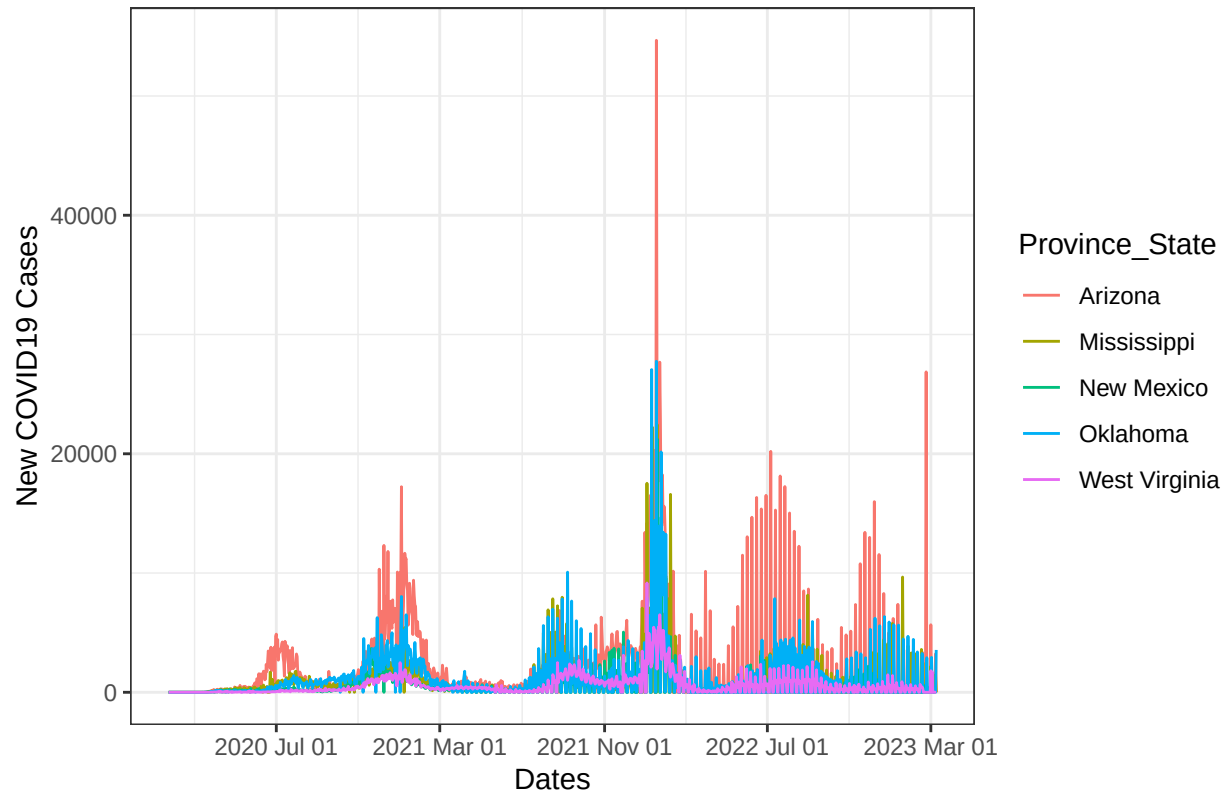


By deaths_per_100, most affected states: Arizona, Oklahoma, Mississippi, West Virginia & New Mexico

```
# Create a new dataframe for Iron County and add columns for daily new cases and deaths.
top_states_deaths <- states_detail_data %>%
  filter(Province_State %in% c("Arizona", "Oklahoma", "Mississippi", "West Virginia", "New Mexico"),
         new_cases >= 0, new_deaths >= 0)

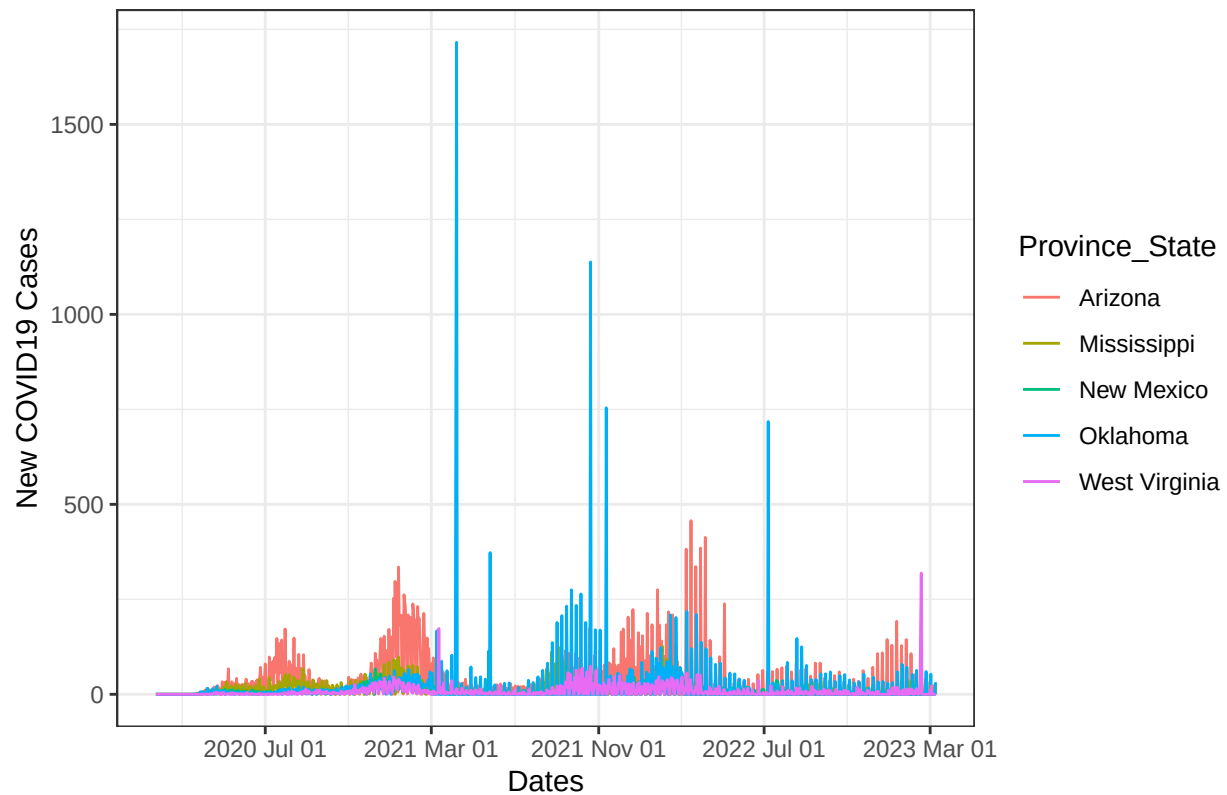
ggplot(top_states_deaths, aes(x=date)) +
  geom_line(aes(y = new_cases, color=Province_State)) +
  scale_x_date(date_labels = "%Y %b %d", date_breaks = "8 month") +
  theme_bw() +
  labs(x = "Dates",
       y = "New COVID19 Cases",
       title = "Most affected states New COVID19 Cases - Time Series")
```

Most affected states New COVID19 Cases – Time Series



```
ggplot(top_states_deaths, aes(x=date)) +
  geom_line(aes(y = new_deaths,color=Province_State)) +
  scale_x_date(date_labels = "%Y %b %d", date_breaks = "8 month") +
  theme_bw() +
  labs(x = "Dates",
       y = "New COVID19 Cases",
       title = "Most affected states New COVID19 Deaths - Time Series")
```

Most affected states New COVID19 Deaths – Time Series



Regression Model

Logistic Regression model was used to predict future Covid-19 cases and deaths in US.

Predictive Variable: deaths_per_100

Dependent Variables: cases_per_100

```
# Build the linear regression model.
lr_model <- lm(deaths_per_100 ~ cases_per_100, data = states_total)

# Display summary for model analysis.
summary(lr_model)
```

```
##
## Call:
## lm(formula = deaths_per_100 ~ cases_per_100, data = states_total)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.22690 -0.05992  0.01527  0.06138  0.12153
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.014972   0.045856  -0.327   0.745
```

```
## cases_per_100 0.010656 0.001494 7.132 2.09e-09 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08471 on 56 degrees of freedom
## Multiple R-squared: 0.476, Adjusted R-squared: 0.4666
## F-statistic: 50.86 on 1 and 56 DF, p-value: 2.095e-09
```

Model Observations

- Coefficients:

coefficients represent the effect of the predictor variables on the response variable. The summary of coefficients is:

Intercept: -0.014972 (Estimate), 0.045856 (Std. Error), -0.327 (t value), 0.745 (Pr(>|t|)) cases_per_100: 0.010656 (Estimate), 0.001494 (Std. Error), 7.132 (t value), 2.09e-09 (Pr(>|t|)) The significance codes indicate that **cases_per_100** is highly significant (denoted by ***).

- R-Squared:

Multiple R-squared: 0.476

Adjusted R-squared: 0.4666

These values indicate the proportion of the variance in the dependent variable that is predictable from the independent variable. An R-squared value of 0.476 implies that approximately 47.6% of the variability in **deaths_per_100** can be explained by the **cases_per_100**.

- F-Statistic

The F-statistic is 50.86, computed on 1 and 56 degrees of freedom, with a p-value of 2.095e-09. This statistic tests whether the overall regression model is a good fit for the data. The very small p-value indicates that the model is highly significant.

- In conclusion, the linear regression model shows a strong and significant correlation between the number of deaths per 100 and the number of cases per 100, with the predictor variable **cases_per_100** having a significant positive effect on the response variable **deaths_per_100**.

Bias Identification

The analysis conducted in this report is not without bias as every source of data is biased in some way. When looking at any results from this report we should consider that each state is different in many ways, including population and health policy, and so the COVID-19 virus affected each state different.

Conclusion

In terms of total cases and deaths, the most affected states were California, Texas, Florida & New York. However when checked by cases_per_100, it seems the most affected states were Rhode Island, Alaska, Kentucky, North Dakota & Guam. Whereas when checked by deaths_per_100, it seems the most affected states were Arizona, Oklahoma, Mississippi, West Virginia & New Mexico.

Also, the linear regression model shows a strong and significant correlation between the number of deaths per 100 and the number of cases per 100, with the predictor variable **cases_per_100** having a significant positive effect on the response variable **deaths_per_100**

In conclusion, this project has been an informative exploration for Covid-19 data for US. Further analysis can be done on Global Covid-19 data, to further extend these observations. Also, additional fields can also be combined in the model to further improve the model and prediction accuracy.